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Unicorn Story Generation and Limits of Words – – On Perspectives of Automatic Tacit Knowledge Addition

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Abstract: In our paper we discuss the problem of tacit knowledge which probably is one of the biggest obstacles on the way to human-level language understanding. While the latest massive transformer-based NLP algorithms show the potential to generate natural text, translate and answer questions, these achievements are still insufficient to directly help machines to acquire common sense. We introduce the problem of tacit knowledge and various approaches to solving it by knowledge completion with automatic text generation which is meant to enrich existing texts and to improve machine learning process by filling the semantic gaps naturally omitted by human beings while generating a natural language. We present various types of knowledge additions we performed in the past and report on the latest achievements in Schankian scripts generation.

1 Introduction

Natural language has become a common input, output and important knowledge source for Artificial Intelligence applications, but more general AI needs more general knowledge and the language only is not the easiest source for learning. When humans process language, their experience-based intelligence and economy of representing thoughts lead to abbreviations [Martinet 64]. This process causes scarce world representations because we do not write and say what we assume understandable *without saying*. Language is meant to convey as much meaning as possible in the possibly shortest form.

1.1 Tacit Knowledge

Already over a century ago Hermann Paul noticed that abundant use of linguistic means of expressing a thought is determined by the need [Paul 88]. Expressions are made in order to contain only just so much as is requisite to their being understood. The amount of linguistic material varies depending on a situation and we perform relative approximations of the receiver's common state of mind. This agrees with Zipf's theories, i.e. the Force of Unification, Speaker's Economy, and correlates with his Principle of Least Effort [Zipf 13]. All of these theories underline simplification in everyday communication [Horn 93] and are also visible in written language. Humans use pronouns (*John runs faster than* ***she*** *does*) and replace whole phrases to avoid repetitions (*Is it going to rain? I think* ***so***). However, even if we complete these sentences with a name (*John runs faster than* ***Mary*** *does*) or repeat the phrase (*Is it going to rain? I think* ***it is going to rain***)), we do not add implicit knowledge [Dienes 99] about e.g. places where running and raining usually takes place, because we assume our interlocutor or reader knows it and / or they are not important for the message. In this paper we call such missing knowledge *tacit knowledge*, but in a slightly different meaning than the definition of [Polanyi 09], who coined the term.

This term was later used differently in various cases [Gourlay 04], but Polanyi concentrated on pieces of knowledge which cannot be (easily or at all) verbalized as in riding a bike. We define the term as a set of assumed facts which are possible to verbalize but which are not verbalized due to the economy of language-related omissions described above. For ex-

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ample when natural language users write about riding a bike they normally do not provide details on the sub-events as sitting on a saddle, starting pedaling, riding outside and having fun assuming that message receivers have this experience in their mental imagery, so senders do not include this information unless the attention must be directed to such detail due to e.g. abnormality.

1.2 Tacit Knowledge and Artificial Intelligence

The above-described phenomenon of language redundancy is one of the main problems causing insufficient knowledge for text-based machine learning which still is not able to derive inferences from commonsensical relations necessary for deeper context understanding. Probably the most natural way to fill the contextual gaps would be to add information from sensors similarly to human perception, but the image [Cheng 18] and video [Aafaq 18] understanding fields are still not capable of high quality recognition of situations and events. Therefore the written text remains the best source for context processing and word embeddings, which preserve more contextual relations, have shown lately higher usability over methods not using distributed representations. However, they still are limited to finding related words without semantic categorization and the relations are vague ($\overrightarrow{woman} + \overrightarrow{king} - \overrightarrow{man} \simeq \overrightarrow{queen}$ relation resemblance is easy to interpret but the human needs to provide a meaningful seed analogy and to deal with noise when less obvious pairs are input or output). This leads to problems with answering basic commonsensical questions and impedes explainability of AI systems. In our research we try to implement methods to solve these problems, at least partially, by artificially adding commonsense knowledge to existing written texts in order to enrich current and future machine learning methods for deeper language understanding. By extending existing text resources we also aim at allowing machines to discover when and how humans abbreviate language – by both omission or replacement (with pronouns, hypernyms, etc.).

2 Related Work

The task of semantic enrichment is not new, however it has gained a new momentum in recent years due to the new enormous text resources and faster computers. Resolving anaphoras [Markert 05], especially in null-subject languages, is probably one of the oldest problems where knowledge addition was performed. However, back then and in most cases now, the clues for additions need to be supervised, e.g. through costly annotations. Most of the work is done on English, but as we work on Japanese, some of the examples in this section regard research involving this language.

Also in the case of common sense knowledge, human was the main source of input [Lenat 89, Niles 01, Liu 04], hence weakly supervised or unsupervised approaches became more desirable for obtaining concepts and their semantic relations. For example [Nastase 08, Krawczyk 16] have shown that Wikipedia categories can be used for extracting implicit relations and [Tandon 11, Shudo 16] have introduced bootstrapping methods for achieving it from raw Web texts. Sentiment information is another example of enriching knowledge for natural language processing. [Cambria 12] presented crowdsourcing method for obtaining affect-related concepts via Games with Purpose but lately the same group introduced a targeted aspect-based sentiment analysis via embedding commonsense knowledge into an attentive LSTM [Ma 18].

The distributed representations have became a new standard for knowledge enrichment. For example Facebook researchers introduced a method for enriching word vectors by using subword information [Bojanowski 17]. Their approach is based on the skipgram model, where each word is represented as a bag of character n-grams. A vector representation is associated to each character n-gram and words are represented as the sum of these representations. More semantics-oriented method was implemented by [Alhussien 18] who have recently showed how to semantically enhance common sense knowledge acquisition. Their models perform joint representation learning from structural and semantic resources and derive a set of semantically salient resources that cover structural, semantic, affective and taxonomical aspects of concepts.

Document enrichment is another closely related task. [Zhang 15] retrieve relevant knowledge from external resources, but in contrast to conventional approach which primarily relies on special resources, they instead use triples of Subject, Predicate and Object as knowledge and incorporate distributional semantics to rank them. Their model first extracts these triples automatically from raw text and converts them into

real-valued vectors based on the word semantics captured by Latent Dirichlet Allocation.

Text generation is another way to deal with missing words [Wang 19] or even longer fragments of text which could be useful for filling omissions due to the language redundancy. Recent successes in applying transformer and attention models [Vaswani 17] to classic NLP tasks [Wu 18] have revealed the potential of powerful computing over massive amounts of raw text data and outperformed previous deep learning methods as recurrent or convolutional neural networks. In February 2019 Open AI has published some effects of their trained large-scale unsupervised language model [Radford 18a] which generates coherent paragraphs of text based on their GPT model [Radford 18b]. It achieves state-of-the-art performance on many language modeling benchmarks, and performs rudimentary reading comprehension, machine translation, question answering, and summarization – all without task-specific training. As a successful example of text generation, an automatically created story about unicorns found in Andes Mountains was given[1]. From a few sentences about a fictional scientist and his discovery, the model elaborated on this topic in few natural paragraphs and the output caused concerns within the company which decided not to release their largest trained model to avoid malicious use of the data, e.g for generating highly believable fake news. However, as reading comprehension example shows, the problems with commonsense knowledge still remains. When a passage about carrying the Olympic torch was given, GPT-2 asked about climbing any mountains on the way answered "Mount Everest" while the name was given in a different context.

To enrich knowledge about events which could be helpful in tracking common acts and avoiding similar errors, several attempts to discover temporal relations have been made, [Hong 18] being one of the latest examples. They differ from usual knowledge base completion tasks which deal with relations between entities, not commonsensical events [Socher 13, Xue 18], although [Sedghi 18] aim at retrieving generic facts as "all trees produce oxygen" which is closer to our goals. Graphs are probably the most popular representation of knowledge, and a plethora of graph-based methods have been introduced. Except classic subtasks as path prediction or relation discovery, also generation and enrichment are widely studied [Guo 17, Hao 17, Ji 18]. However, methods for finding relations as these mentioned above or ones we introduced [Krawczyk 16], use already existing knowledge bases as Wikipedia which is meant to enrich human expertise and not to augment machine's common sense. For such knowledge ontologies as ConceptNet [Liu 04] were developed, however languages other than English are underrepresented and before working on graphs, we decided to mostly work on knowledge discovery from raw texts and extend them by adding missing information to the analyzed sentences. Such expansion was not reasonable until recently and the graph representation is easier to maintain, however we feel that the knowledge addition might be better in preserving contextual nuances.

Due to the space constraints of a short paper, numerous related works were not mentioned [Kobayashi 18], but we assume that [Bai 17] and Data Programming [Ratner 16] may represent approaches closest to ours. The first research tries to implement various cognitive theories in order to discover implicit knowledge categorized by conditionality, causality, concession and other relational schemas. The latter describes a wider paradigm of automatic preparation of learning data, and we believe that combining both could lead to richer knowledge representations unavailable for current machine learning systems.

3 Unzipping Context for AGI – Past Examples

Artificial General Intelligence is a term containing a set of problems in the process of achieving human-level capabilities in machines. We treat commonsense knowledge and commonsense reasoning as the main components for trustful, safe or even creative AGI and we try to acquire knowledge for world simulation in an unsupervised or weakly-supervised way. This is because annotating such knowledge is difficult due to the discrepancies between annotators whose experiences and personalities vary – what is common for one person in context X, can be strange for another person and generality values can be different in context Y. It can be said that we regard the collective wisdom of the Web data as a supervisor in inverse reinforcement learning [Ng 00]. Humans indirectly show our systems how to react or not react and what are their usual reactions to a given act. Our basic idea was a swarm of life-long learning web-crawlers

[1] https://openai.com/blog/better-language-models/

[Rzepka 03a] that scrap different parts of the Internet and share their findings as Minskian critics. For example, enriching the user's input with discovered possible emotional reactions and the level of generality helped to achieve more natural utterances in dialog systems [Rzepka 03b, Rzepka 05b]. We used this approach also in automatic moral assessment of acts [Rzepka 05a], and tackled yet another example of "unspoken" or "unwritten" knowledge problem, i.e. pragmatical side of a language where "can you pass the salt?" is not a question about your capability to perform this act, but a request [Grice 75]. We dealt with this problem in a robotic vacuum-cleaner experiments and showed that automatic knowledge addition to users' utterances allows to interpret utterances like "this room is a mess" as a signal to the robot that its help might be needed [Takagi 11]. By finding information about other similar robots, the system could also predict what types of orders are impossible to be performed. However, the cleaning task is quite narrow and more general knowledge is necessary to adapt to more diverse world scenarios. We also simulated five senses by retrieving related descriptive words as adjectives and onomatopoeias [Rzepka 13] and automatically generated assertions for concepts like $LocationOf$, $CreatedBy$, $MadeBy$, or $UsedFor$ [Krawczyk 16, Shudo 16]. Nevertheless, we have failed to generate chains of acts longer than one pair of event [Ge 05]. Almost 15 years later, we have managed to break this limitation by using bigger resources and word embeddings.

4 Extending Schankian Scripts

More than 40 years ago [Schank 77] introduced the concepts of scripts, plans and themes to handle story-level understanding. Most famous "restaurant script" consists of four scenes (Entering, Ordering, Eating, Exiting) which contain more detailed information about an act (e.g. $\mathcal{S}$ *PTRANS* $\mathcal{S}$ into restaurant, $\mathcal{S}$ *ATTEND* eyes to tables, $\mathcal{S}$ *MBUILD*< where to sit, $\mathcal{S}$ *PTRANS* $\mathcal{S}$ to table, $\mathcal{S}$ *MOVE* $\mathcal{S}$ to sitting position, etc.) In spite of various trials, single causal relation as "sit down → look at menu" have not been successfully extended automatically because humans rarely write what we exactly do and, even more naturally, what they do not do. We have used two Japanese Web corpora: YACIS [Ptaszynski 12] (paragraphs processed) and NWC[2] (single sentences processed). For example, when analyzed sentence was シャンプーで洗髪した後にお湯で洗う (*After shampooing (my) hair, (I) washed in with hot water*), keyword analysis and semantic parsing use the temporal marker "after" (後に) to create an event pair in the discovered order: *shampooing hair* THEN *washing with hot water*. Acts extracted from a noisy raw corpus naturally include semantically irrelevant data, therefore we calculate relevance between the events distributed word representations and acquire only highly (above experimentally set threshold) related events. The relevance calculation is shown in Equation (1), where W_{Ak} is the kth word of word set A, W_{Bn} is the nth word of word set B and N is the total of words in A.

$$similarity(W_{Ak}) = \frac{\sum_{n=0}^{N-1} cosine(W_{Ak}, W_{Bn})}{N} \quad (1)$$

In the next step event pairs are combined and event chain candidates are generated. By extracting events from the whole corpora the system managed to acquire implicit event candidates. When event B of event pair A→B and event C of event pair C→D were highly relevant, event B and event C were considered likely to occur in the same situation, and to usually appear one after another. The semantic relevance between each event pair and all remaining ones is calculated with Equation (2), and M is the total of words in B and again thresholds are used.

$$similarity(Event_A, Event_B) = \frac{\sum_{k=0}^{K-1} \sum_{n=0}^{N-1} cosine(W_{Ak}, W_{Bn})}{MN} \quad (2)$$

This threshold calculation method and other details of this method are given in [Takishita 19]. A single event is limited to verb phrase containing an object, but other information as actor, place or tool are

Table 1: Third-person evaluation of event pairs depending on length (3 or 4) and used corpus.

	Parts					
Corpus	AB	AC	AD	BC	BD	CD
YACIS-3	2.8	2.5	-	2.6	-	-
YACIS-4	2.8	2.4	2.5	2.4	2.5	2.7
NWC-3	2.8	2.0	-	2.0	-	-
NWC-4	2.8	1.7	1.9	1.8	2.0	2.6
Average	2.8	2.15	2.2	2.2	2.25	2.65

[2] http://s-yata.jp/corpus/

omitted. We are currently experimenting with adding these elements to the process. Examples of generated 4-event chains are shown below:

- 頭皮をよく洗う → シャンプーで洗う → お湯をかき混ぜる → 入浴する (*washing head well → washing with shampoo → stirring hot water → entering bath-tube*)
- 記事を書く → ブログを読む → 本を読む → 英語を勉強する (*writing article → reading blog → reading book → learning English*)
- お金を払う → お店を借りる → 営業を営む → 宣伝を行う (*paying money → renting shop → conducting business → making advertisment*)

The third-person evaluation for 3- and 4-event chains results are shown in Table 1.

5 Conclusions and Future Work

The main goal of this paper is to introduce the problem of tacit knowledge to the AGI community. After presenting examples of various approaches of other researchers and of our own, we briefly present the latest method for generating longer event chains which are an important element in acquiring richer behavior patterns for context understanding. We propose a novel approach to add tacit knowledge directly to input sentences which should facilitate natural language understanding and, by allowing better explainability, also lead to more trustful AGI. As the feature work, except adding semantic roles, we plan to start combining all previous additions into one vector space and empirically confirm the validity of such combinations in Winograd Schema Challenge [Levesque 12] and our other context-dependent projects like moral judgement, dialog understanding or metaphor processing.

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